Status	Finished
Started	Monday, 19 October 2026, 5:03 PM
Completed	Monday, 19 October 2026, 5:31 PM
Duration	28 mins 26 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

What is searching in the context of computer science?

a. Inserting elements into a list
b. Determining whether an element is present in a list ✓
c. Deleting elements from a list
d. Sorting elements in a list

Question **2**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

a. When the list contains duplicate elements
b. When the list is sorted
c. When the element is found in the list ✓
d. When the search algorithm finishes

Question **3**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

a. O(1) ✓
b. O👎
c. O(n log n)
d. O(log n)

Question **4**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

a. Binary search works on unsorted lists
b. Binary search does not require dividing the list
c. Binary search has a lower time complexity on large, sorted lists ✓
d. Binary search can find multiple instances of the target element

Question **5**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

a. 1
b. n/2
c. log n
d. n ✓

Question **6**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

a. It requires the list to be sorted ✓
b. It can only be applied to large lists
c. It does not work with negative numbers
d. It is slower than linear search for small lists

Question **7**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

a. There must be a mechanism to delete and/or insert elements in the list ✓
b. Number values should only be present

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c. There should be direct access to the middle element in any sublist
d. The list must be sorted

Question **8**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

- a. Linear search ✓
- b. Both (1) & (3)
- c. Hash search
- d. Binary search

Question **9**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

- a. O👎 ✓
- b. O(log n)
- c. O(n log n)
- d. O(1)

Question **10**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

- a. 1000
- b. 100
- c. 10 ✓
- d. 500

Question **11**

Correct

Mark 1.00 out of 1.00

Which method of searching involves sequentially comparing each element until a match is found?

- a. Binary search
- b. Jump search
- c. Hashing
- d. Linear search ✓

Question **12**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

a. By sorting the list first
b. By using a hash function
c. By comparing each element in the list sequentially ✓
d. By dividing the list into halves

Question **13**

Correct

Mark 1.00 out of 1.00

_____sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements in case they are unordered in n-1 passes.

a. Complexity

b. Insertion

c. Selection

d. Bubble ✓

Question **14**

Correct

Mark 1.00 out of 1.00

What happens when the element is found in linear search?

a. The search backtracks to find duplicate elements
b. The search stops immediately ✓
c. The search starts over from the beginning
d. The search continues until the end of the list

Question **15**

Correct

Mark 1.00 out of 1.00

What is the key characteristic of binary search?

a. It can be applied only if the list is sorted ✓
b. It works on unsorted lists
c. It always starts from the beginning of the list
d. It compares elements sequentially